Maximizing Local Entropy Where It Matters: Prefix-Aware Localized LLM Unlearning

在关键区域最大化局部熵：基于前缀感知的局部 LLM 去学习

Naixin Zhai [1] , Pengyang Shao [2,*] ,

纳新斋 [1] ，彭阳邵 [2,*] ，

Binbin Zheng [1] , Yonghui Yang [2] , Fei Shen [2] , Long Bai [2] , Xun Yang [1,*]

郑彬彬 [1] ，杨永辉 [2] ，沈斐 [2] ，白龙 [2] ，杨迅 [1,*]

[1] University of Science and Technology of China

[1] 中国科学技术大学

[2] National University of Singapore

[2] 新加坡国立大学

zhainaixin@mail.ustc.edu.cn, shaopymark@gmail.com, xyang21@ustc.edu.cn

zhainaixin@mail.ustc.edu.cn, shaopymark@gmail.com, xyang21@ustc.edu.cn

Abstract

摘要

Machine unlearning aims to forget sensitive knowledge from Large Language Models (LLMs) while maintaining general utility. However, existing approaches typically treat all tokens in a response indiscriminately and enforce uncertainty over the entire vocabulary. This global treatment results in unnecessary utility degradation and extends optimization to content-agnostic regions. To address these limitations, we propose PALU (Prefix-Aware Localized Unlearning), a framework driven by a local entropy maximization objective across both temporal and vocabulary dimensions. PALU reveals that (i) suppressing the sensitive prefix alone is sufficient to sever the causal generation link, and (ii) flattening only the top-K logits is adequate to maximize uncertainty in the critical subspace. These findings allow PALU to alleviate redundant optimization across the full vocabulary and parameter space while minimizing collateral damage to general model performance. Comprehensive evaluations validate that PALU achieves superior forgetting efficacy and utility preservation compared to state-of-the-art baselines. Our code is available at https://github.com/nxZhai/PALU.

机器遗忘旨在在保持通用实用性的同时，使大型语言模型（LLMs）忘记敏感知识。然而，现有方法通常对响应中的所有标记进行无差别处理，并在整个词汇表上强制不确定性。这种全局处理导致了不必要的实用性下降，并将优化扩展到与内容无关的区域。为了解决这些限制，我们提出了 PALU（Prefix-Aware Localized Unlearning），一个基于时间维度和词汇维度上局部熵最大化目标的框架。PALU 揭示了以下两点：(i) 仅抑制敏感前缀就足以切断因果生成链接，(ii) 仅平坦化前 K 个 logits 就足以在关键子空间中最大化不确定性。这些发现使 PALU 能够在不损害通用模型性能的情况下，减少对整个词汇表和参数空间的冗余优化。全面的评估验证了 PALU 在遗忘效果和实用性保持方面优于最先进的基线方法。我们的代码可在 https://github.com/nxZhai/PALU 获取。

1 Introduction

1 引言

Large language models have been widely deployed across diverse domains (Hu et al., 2024; Xu et al., 2025; Han et al., 2025), yet they inevitably memorize sensitive, private, and copyrighted information from massive training corpora (Luo et al., 2025; Li et al., 2025; Fang et al., 2025; Zhu et al., 2026). Such memorization not only raises security and ethical concerns (Karamolegkou et al., 2023; Zhao et al., 2024b,a; Pan et al., 2025), but also conflicts with data privacy regulations such as GDPR [1]

大型语言模型已被广泛部署在各种领域（Hu 等，2024；Xu 等，2025；Han 等，2025），然而它们不可避免地会从庞大的训练语料库中记忆敏感、私密和受版权保护的信息（Luo 等，2025；Li 等，2025；Fang 等，2025；Zhu 等，2026）。这种记忆不仅引发了安全和伦理方面的担忧（Karamolegkou 等，2023；Zhao 等，2024b,a；Pan 等，2025），还与诸如 GDPR 等数据隐私法规相冲突 [1]

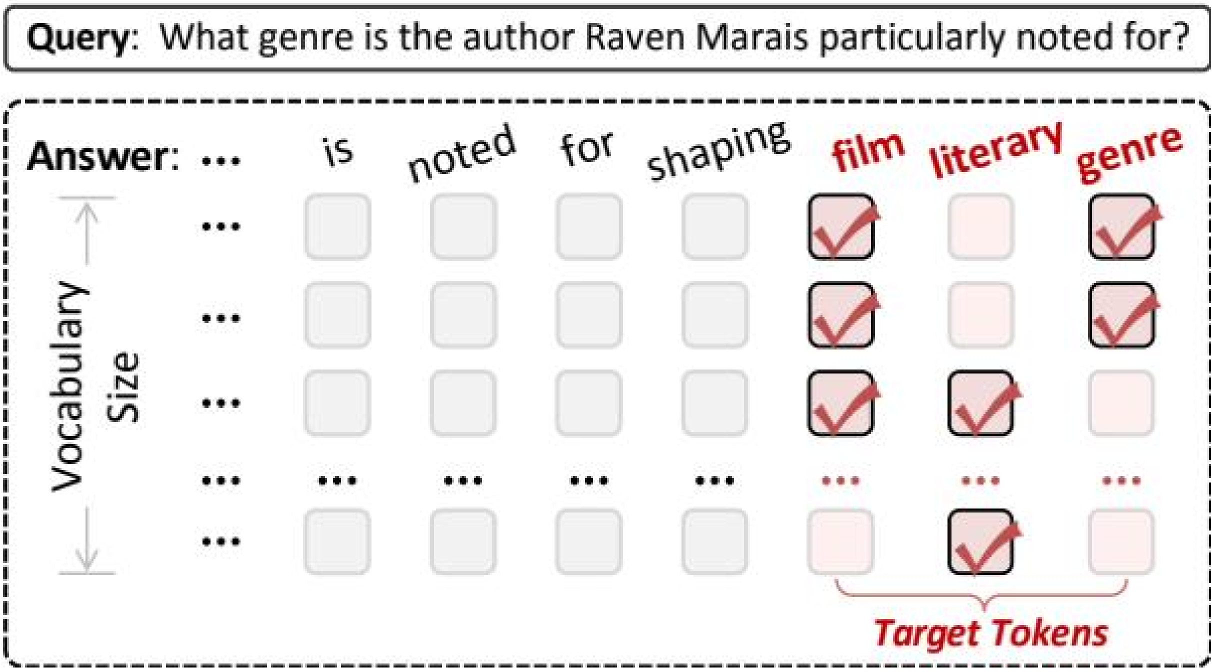


Figure 1: images/3f32b8292c39ce17596d9ae542791cd215239cf89d3cc93e5868674ca235199d.jpg

Figure 1: Illustration of the vocabulary-localized optimization. We specifically target sensitive tokens (red) while bypassing context-agnostic ones (gray). For each target position, the optimization is restricted to the top-K vocabulary candidates (indicated by □), thereby pruning the computation on long-tail dimensions.

图 1：词汇本地化优化的示意图。我们特别针对敏感词（红色）进行优化，而绕过上下文无关的词汇（灰色）。对于每个目标位置，优化仅限制在前 K 个词汇候选（用 □ 表示），从而剪枝长尾维度的计算。

and CCPA [2] , which grant individuals the “right to be forgotten”. Consequently, machine unlearning, which selectively removes targeted information from trained models without retraining from scratch, has emerged as a prerequisite for the safe and compliant deployment of LLMs (Yao et al., 2024; Tirumala et al., 2022; Goel et al., 2026).

以及《加州消费者隐私法案》（CCPA）[2]，该法案赋予个人 “被遗忘权”。因此，机器反学习（machine unlearning）作为一种无需从头重新训练即可选择性地从训练模型中移除特定信息的技术，已成为大型语言模型（LLMs）安全合规部署的必要前提（Yao 等，2024；Tirumala 等，2022；Goel 等，2026）。

Despite growing progress, current LLM unlearning methods remain largely grounded in variants of the negated cross-entropy (CE) objective. Representative approaches, such as GradientAscent (GA) (Yao et al., 2024) and Negative Preference Optimization (NPO) (Zhang et al., 2024), primarily aim to suppress the probability of the top-1 token. While intuitive, negated CE often leads to over-correction compared to entropy maximization, which naturally promotes uniform uncertainty without aggressively destroying contextual knowledge (Entesari et al., 2025; Pan et al., 2024; Zhao et al., 2025). Beyond this objective-level limitation, these methods share a common structural drawback: they induce global interventions by applying dense gradients across the full response

尽管取得了显著进展，当前的 LLM 去学习方法仍然主要基于否定交叉熵（CE）目标的各种变体。代表性方法，如 GradientAscent（GA）（Yao 等，2024）和 Negative Preference Optimization（NPO）（Zhang 等，2024），主要旨在抑制 top-1 token 的概率。虽然这种做法直观，但与熵最大化相比，否定 CE 常常导致过度校正，而熵最大化自然地促进均匀的不确定性，而不会激进地破坏上下文知识（Entesari 等，2025；Pan 等，2024；Zhao 等，2025）。除了这一目标层面的限制外，这些方法还有一个共同的结构缺陷：它们通过在整个响应上应用密集梯度来诱导全局干预。

sequence and a large portion of the vocabulary. This global reshaping can inadvertently suppress content-agnostic functional words, disrupt linguistic coherence, and degrade general utility (Chen and Yang, 2023; Ji et al., 2024). It also incurs substantial computational overhead, as optimization must backpropagate through all token positions and vocabulary dimensions, irrespective of their relevance to the sensitive content.

序列和大量词汇。这种全局重塑可能会无意中压制内容无关的功能词，破坏语言的连贯性，并降低通用性（Chen 和 Yang，2023；Ji 等，2024）。此外，它还会带来显著的计算开销，因为优化必须通过所有 token 位置和词汇维度进行反向传播，无论这些位置和维度是否与敏感内容相关。

In this work, we revisit LLM unlearning through the lens of intervention efficiency: achieving effective forgetting with the minimal necessary perturbation to model parameters. From this perspective, unlearning can be viewed as disrupting the generation trajectory that leads from a query x to an undesired response y. Crucially, such disruption need not be global, motivating two key observations. (i) Temporal Sparsity: Sensitive semantics are typically triggered by a small prefix of pivotal tokens. Intervening on this initiating prefix is often sufficient to divert the generation path, making updates to subsequent tokens redundant. (ii) Vocabulary Sparsity: In most memorization scenarios, autoregressive decoding decisions are dominated by a small set of high-probability candidates, rendering interventions on long-tail vocabulary dimensions largely unnecessary. Flattening only the dominant logits can already induce strong confusion and divert the decoding path.

在本工作中，我们从干预效率的角度重新审视大语言模型（LLM）的遗忘问题：通过最小必要扰动模型参数来实现有效的遗忘。从这一角度来看，遗忘可以被视为干扰从查询 $ x $ 到不期望响应 $ y $ 的生成轨迹。关键的是，这种干扰并不需要是全局的，这促使我们得出两个重要的观察结论。(i) 时间稀疏性：敏感语义通常由一小段关键的标记触发。对这一初始前缀进行干预通常就足以改变生成路径，使得后续标记的更新变得冗余。(ii) 词汇稀疏性：在大多数记忆场景中，自回归解码决策通常由一小部分高概率候选词主导，使得对长尾词汇维度的干预基本上是不必要的。仅对主导的 logits 进行扁平化处理，就已经能够引发强烈的混淆并改变解码路径。

Guided by these insights, we propose PALU, a cost-efficient unlearning framework built upon a dual-sided localized entropy maximization objective, localized in both time and vocabulary. PALU first efficiently identifies a sensitive decoding prefix. For these selected positions, it then applies a localized loss restricted to the top-K logits, as illustrated in Figure 1. This targeted intervention maximizes predictive uncertainty within the decoding-critical subspace, effectively steering generation away from the sensitive trajectory while pruning redundant updates on irrelevant tokens and long-tail vocabulary dimensions. Extensive experiments show that PALU achieves state-of-the-art forgetting efficacy while significantly improving utility preservation compared to strong baselines. Our contributions are summarized as follows:

基于这些见解，我们提出了 PALU，这是一个基于双侧局部熵最大化目标的高效去学习框架，该目标在时间和词汇上都具有局部性。PALU 首先高效地识别出敏感的解码前缀。对于这些选定的位置，它随后应用一个局限于前 K 个 logits 的局部损失，如图 1 所示。这种有针对性的干预在解码关键子空间内最大化预

测不确定性，有效引导生成远离敏感轨迹，同时修剪与无关标记和长尾词汇维度相关的冗余更新。大量实验表明，PALU 在实现遗忘效果方面达到了最先进的水平，同时在与强大基线相比显著提高了效用保留。我们的贡献总结如下：

(1) We revisit LLM unlearning through the lens of intervention efficiency and formulate it as disrupting the sensitive generation trajectory with minimal necessary intervention.
(2) We propose PALU, which introduces a dual-sided localized entropy maximization objective,

(1) 我们从干预效率的角度重新审视大语言模型的遗忘问题，并将其表述为以最小必要干预的方式扰乱敏感生成轨迹。
(2) 我们提出 PALU，它引入了一个双侧局部熵最大化的目标函数，

thereby reducing redundant computation and collateral degradation.

从而减少冗余计算和附带的退化。

(3) Empirical results demonstrate that PALU sets a new standard for the trade-off between forgetting quality and utility preservation.

(3) 实证结果表明，PALU 在遗忘质量与效用保持之间的权衡设定了新的标准。

2 Related Work

2 相关工作

2.1 LLM Unlearning

2.1 LLM 去学习

LLM unlearning focuses on removing targeted knowledge from LLMs while preserving general utility. Early methods such as GradientAscent (GA) and GradientDiff (GD) (Maini et al., 2024) maximize the loss on forget samples, typically via the negated CE. However, such unbounded objectives often cause unstable updates, e.g., excessive probability suppression and over-refusal during generation (Zhang et al., 2024). NPO (Zhang et al., 2024) introduces bounded objectives anchored to a reference model; SimNPO (Fan et al., 2025) removes reference-model bias for efficiency, and AltPO (Mekala et al., 2025) incorporates positive feedback to reduce nonsensical refusals. Further studies improve the trade-off between stability and utility by better refining negated CE, e.g., token saturation reweighting (SatImp) (Yang et al., 2025).

LLM 去学习聚焦于从 LLM 中移除特定知识，同时保持其通用的实用性。早期的方法如 GradientAscent（GA）和 GradientDiff（GD）（Maini 等，2024）通过否定交叉熵（CE）最大化遗忘样本的损失。然而，这种无界的优化目标常常导致不稳定的更新，例如在生成过程中出现过度的概率抑制和过度拒绝（Zhang 等，2024）。NPO（Zhang 等，2024）引入了以参考模型为基准的有界优化目标；SimNPO（Fan 等，2025）为了提高效率去除了参考模型的偏差，而 AltPO（Mekala 等，2025）则通过引入正面反馈来减少无意义的拒绝。进一步的研究通过更好地细化否定交叉熵，改进了稳定性与实用性的平衡，例如令牌饱和重加权（SatImp）（Yang 等，2025）。

Recently, token-level LLM unlearning, which intervenes on a subset of tokens instead of suppressing entire sequences, has been widely studied (Wang et al., 2025a; Liu et al., 2025a), with examples including Selective Unlearning (SU) (Wan et al., 2025) and Targeted Preference Optimization (TPO) (Zhou et al., 2026). These methods share a common goal: minimizing unnecessary perturbations for unlearning. However, while these approaches achieve temporal sparsity, they typically overlook redundancy in the vocabulary space. By still computing gradients across the entire output distribution, they continue to rely on expensive dense updates and inherit the limitations of the negated CE objective.

最近，针对 token 级别的 LLM 去学习方法（token-level LLM unlearning）受到了广泛关注，这类方法通过干预部分 token 而非整个序列来实现去学习（Wang et al., 2025a; Liu et al., 2025a），其中包括 Selective Unlearning (SU) (Wan et al., 2025) 和 Targeted Preference Optimization (TPO) (Zhou et al., 2026) 等实例。这些方法有一个共同的目标：在去学习过程中最小化不必要的扰动。然而，尽管这些方法实现了时间上的稀疏性，但它们通常忽略了词汇空间中的冗余。通过仍然在整个输出分布上计算梯度，它们继续依赖昂贵的密集更新，并继承了否定 CE 目标函数的局限性。

2.2 Entropy for Machine Learning

2.2 机器学习中的熵

Entropy has been widely adopted as a principled objective for controlling model behavior in machine learning. The maximum entropy principle advocates selecting the least-committal distribution subject to constraints. This principle motivates entropy maximization as a generic learning objective (Jaynes, 1957; Berger et al., 1996; Wang et al., 2025b; Liu et al., 2026). Prominent applications include entropy regularization to discourage

熵已被广泛采用作为控制机器学习模型行为的有原则的目标。最大熵原理倡导在约束条件下选择最不偏倚的分布。这一原理促使将熵最大化作为通用的学习目标（Jaynes, 1957; Berger et al., 1996; Wang et al., 2025b; Liu et al., 2026）。显著的应用包括使用熵正则化来抑制

over-confident predictions and improve generalization (Jiang et al., 2025), and entropy maximization for reinforcement learning to encourage exploration and robustness (Chao et al., 2024; Cheng et al., 2026; Fu et al., 2026; Zhang et al., 2025a). However, maximizing entropy over the full output space is not always necessary. In many problems, the final prediction is dominated by a small subset of high-probability candidates, while long-tail dimensions contribute little (Gao et al., 2019; Holtzman et al., 2019). This motivates localized entropy maximization, which increases uncertainty only within the prediction-critical subspace (e.g., top-ranked candidates) (Michaud et al., 2023; Entesari et al., 2025), without enforcing global distributional flattening. Building on this intuition, we propose PALU, which leverages localized entropy maximization to achieve intervention efficiency in unlearning, thereby offering a robust alternative to standard suppressive objectives. Unlike negated CE, which purely suppresses probability, entropy maximization naturally induces uniform uncertainty, allowing us to erase knowledge with minimal necessary perturbation.

过度自信的预测和提高泛化能力（Jiang 等，2025），以及通过熵最大化来鼓励探索和鲁棒性的强化学习（Chao 等，2024；Cheng 等，2026；Fu 等，2026；Zhang 等，2025a）。然而，在完整的输出空间上最大化熵并不总是必要的。在许多问题中，最终的预测主要由一小部分高概率候选项主导，而长尾维度的贡献很小（Gao 等，2019；Holtzman 等，2019）。这促使我们进行局部熵最大化，仅在预测关键的子空间（例如排名靠前的候选项）内增加不确定性（Michaud 等，2023；Entesari 等，2025），而不需要强制全局分布平坦化。基于这一直觉，我们提出了 PALU，利用局部熵最大化来实现去学习中的干预效率，从而提供了一种比标准抑制目标更稳健的替代方案。与单纯的抑制交叉熵（negated CE）不同，熵最大化自然地诱导出均匀的不确定性，使我们能够以最小的必要扰动来擦除知识。

3 Preliminary

3 初步准备

Most existing gradient-based LLM unlearning methods can be formulated as optimizing two conflicting objectives (Yao et al., 2024; Si et al., 2023; Shao et al., 2026):

大多数现有的基于梯度的 LLM 去学习方法可以表述为优化两个相互冲突的目标（Yao 等，2024；Si 等，2023；Shao 等，2026）：

$$\min_{\theta} \mathcal{L}_{\text{all}} = \mathcal{L}_f + \lambda \mathcal{L}_r, \tag{1}$$

where the retain loss L_r is typically instantiated as the standard CE loss for next-token prediction on the retain set D_r . In contrast, the forget objective L_f aims to disrupt the learned mapping from the input x to the target output y on the forget set by explicitly suppressing the likelihood of generating y conditioned on x. Optimizing L_f may introduce unnecessary perturbations to model parameters, potentially degrading the model's general capabilities, while L_r serves as an indirect constraint that mitigates such degradation by maintaining performance on the retain distribution. A common instantiation of L_f is the negated CE objective (Liu et al., 2025b; Zhang et al., 2024), defined as:

其中，保留损失 L_r 通常被实例化为在保留集 D_r 上进行下一个 token 预测的标准交叉熵（CE）损失。相反，遗忘目标 L_f 的目的是通过显式地抑制在给定输入 x 的条件下生成目标输出 y 的可能性，从而破坏在遗忘集上学习到的从输入 x 到目标输出 y 的映射。优化 L_f 可能会引入不必要的扰动到模型参数中，从而潜在地降低模型的通用能力，而 L_r 则作为间接约束，通过保持在保留分布上的性能来缓解这种退化。L_f 的常见实例化形式是负的 CE 目标（Liu 等，2025b；Zhang 等，2024），定义为：

$$\mathcal{L}_f = \mathbb{E}_{(x,y)\sim\mathcal{D}_f} \left[\sum_{t=1}^{T} \log p(y_t \mid x, y_{<t}) \right], \tag{2}$$

where T denotes the length of y, and D_f represents the forget set. Despite their different implementation pathways, these methods share a common objective: achieving the desired unlearning while inducing fewer perturbations to LLMs.

其中，T 表示 y 的长度，D_f 表示遗忘集合。尽管这些方法在实现路径上有所不同，但它们共享一个共同的目标：在实现期望的去学习（unlearning）的同时，对大语言模型（LLMs）造成更少的扰动。

In this work, we revisit LLM unlearning from the perspective of unlearning cost, defined as achieving effective unlearning with minimal necessary perturbations to model parameters. Under this perspective, most existing gradient-based methods realize the unlearning objective in Equation 2 through the negated CE loss and its variants. This raises a fundamental question: is negated CE an ideal objective for guiding unlearning?

在本工作中，我们从 “去学习成本” 的角度重新审视大语言模型的去学习问题，其中 “去学习成本” 被定义为通过最小必要地扰动模型参数，实现有效的去学习。从这一角度来看，大多数现有的基于梯度的方法通过否定交叉熵（CE）损失及其变体，实现了方程 2 中的去学习目标。这引发了一个根本性的问题：否定 CE 损失是否是引导去学习的理想目标？

4 The Proposed Method: PALU

4 提出的方法：PALU

4.1 Overview

4.1 概述

We present a cost-efficient unlearning framework, shown in Figure 2, that reduces intervention overhead from two complementary perspectives: the token level and the vocabulary level.

我们提出了一种高效的成本降低框架，如图 2 所示，该框架从两个互补的角度减少干预开销：token 级别和词汇级别。

At the token level, we differentiate optimization targets based on token semantics. Instead of uniformly enforcing unlearning, we selectively apply the unlearning objective to a sparse subset of important initiating tokens, while imposing preservation constraints on the remaining content-agnostic tokens to protect general model capabilities. At the vocabulary level, we reconsider the unlearning objective function. Standard approaches typically employ negative CE for its efficiency, yet this objective often suffers from instability due to naive suppression (Jia et al., 2024). While the global entropy maximization method PDU (Entesari et al., 2025) offers a theoretically superior target for effective erasure, it incurs prohibitive computational costs by optimizing the entire vocabulary space. To reconcile this, we propose local entropy maximization. By selectively flattening only the dominant logits, our method achieves the structural robustness of entropy maximization without the overhead of processing irrelevant long-tail tokens. We detail these formulations in the following subsections.

在词元级别上，我们根据词元语义来区分优化目标。与统一地执行遗忘不同，我们选择性地将遗忘目标应用于一小部分重要的初始词元，同时对剩余的与内容无关的词元施加保留约束，以保护通用模型能力。在词汇表级别上，我们重新考虑遗忘目标函数。标准方法通常采用负的交叉熵（negative CE）以提高效率，但这种目标由于简单的抑制方法（Jia 等，2024）常常导致不稳定。虽然全局熵最大化方法 PDU（Entesari 等，2025）提供了理论上更优的遗忘目标，但其通过优化整个词汇空间而带来了高昂的计算成本。为了解决这一问题，我们提出了局部熵最大化方法。通过选择性地仅展平主导的 logits，我们的方法在不处理理无关的长尾词元所带来的额外开销的情况下，实现了熵最大化方法的结构鲁棒性。我们将在以下子部分中详细阐述这些公式。

4.2 Unlearning via Sparse Initiating Tokens

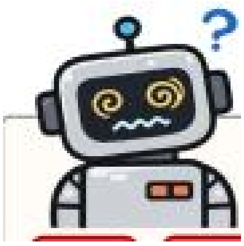
4.2 通过稀疏初始标记进行去学习

We revisit Equation 2 from the perspective of its summation over output tokens. In realistic generation scenarios, output tokens contribute unevenly to sensitive content: many tokens are largely content-agnostic and serve syntactic or stylistic roles, while only a subset of important tokens introduces sensitive semantics (Zhou et al., 2026; Wan et al., 2025). Uniformly enforcing unlearning over all tokens

我们从输出标记的求和角度重新审视方程 2。在现实生成场景中，输出标记对敏感内容的贡献是不均衡的：许多标记在内容上相对中立，主要起到句法或风格的作用，而只有少数重要的标记引入了敏感语义（Zhou 等，2026；Wan 等，2025）。对所有标记统一执行去学习可能会导致问题。

Token-level Unlearning

逐标记的反向学习



He
noted
for
shaping
film
literary
genre

他
指出
在
塑造
电影
文学
类型

1) Classic Gradient-based Methods

1) 经典的基于梯度的方法

Equally treating all tokens in the answer

平等对待答案中的所有标记

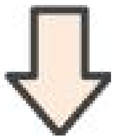
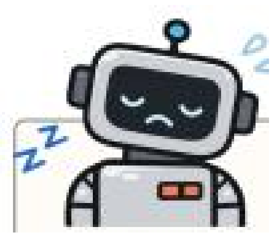


Figure 2: images/a37a6622f65bd1a0a948a936e7c83990c52101cdb3bf70a5a6280081b08121d0.jpg



He
is
noted
for
shaping
film
literary
genre
Common tokens
Important Tokens

他
被認為
塑造
電影
文學
類型
常用標籤
重要標籤

2) Current Token-aware Methods

2) 当前的 Token-aware 方法

Only focus on important tokens

只关注重要的标记

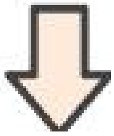
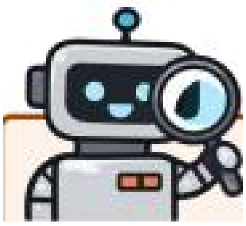


Figure 3: images/641d289bd36aa888d34cc4439da154cc6202954bf49e278973cd1f0c6ff085da.jpg



He
is
noted
for
shaping
film
literary
genre
Common tokens
Initiating token
Redundant Sensitive token

他
以
塑造
电影
文学
类型
常见标记
起始标记
冗余敏感标记

3) Unlearning via Sparse Initiating Tokens

3) 通过稀疏初始化标记进行去学习

□ Only focus on initiating important tokens

□ 只关注启动重要令牌

Vocabulary-level Unlearning

词汇级遗忘

Optimization objectives for the next-token film prediction:

下一个词电影预测的优化目标：

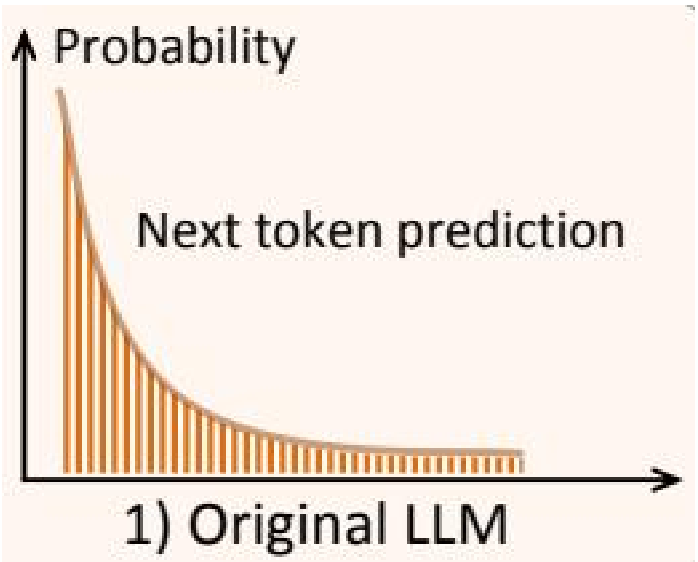


Figure 4: images/5d609d4660701505d7c3b36353e8108bcede0247b75fd3730f14c781694a4c36.jpg

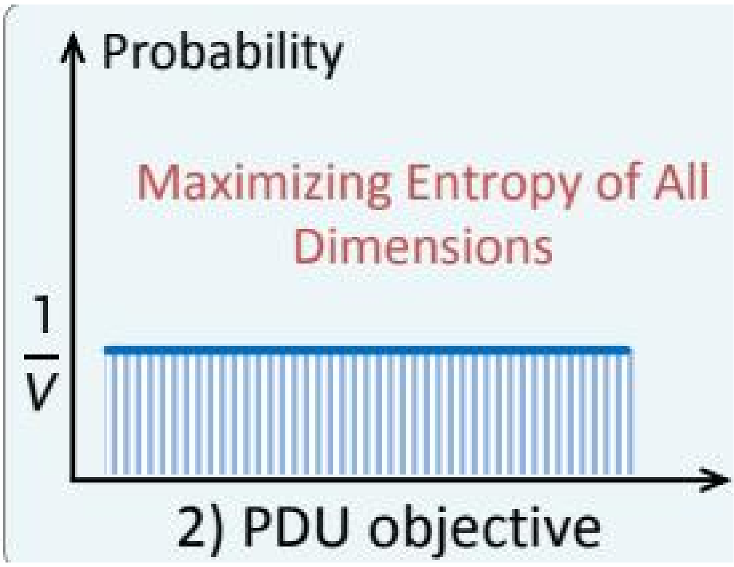


Figure 5: images/8bfa99e95ac4eb8607df8cb0f1da331460b072ad7d70a0a70d55b5b2d432e4bc.jpg

Figure 6: images/54c3955fe4bdf5fe6218fc515e86a7fa9863628fc8a20f77a717b69cc67ecc87.jpg

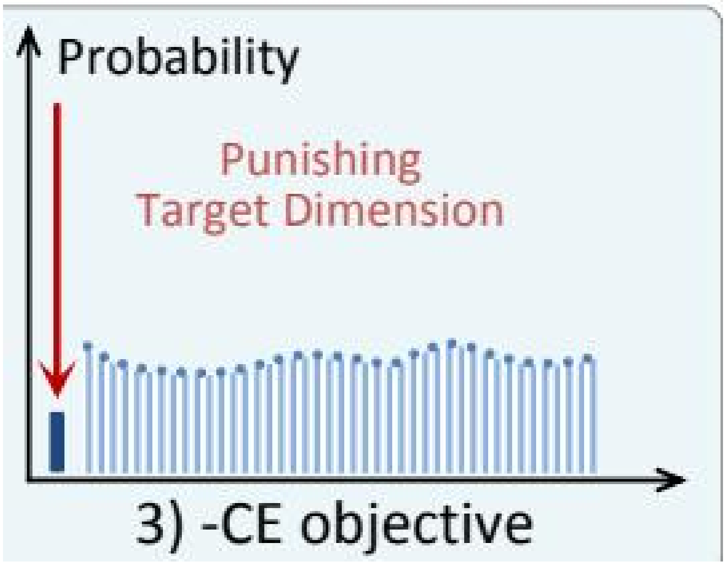


Figure 7: images/928dcc044f0cb094d75f67933dae62a18230b8ccb1f17ecf4b90748bbcc48d96.jpg

Figure 8: images/226e2838c1cab121d7d3e7c734ff3c5efc9147b13351ef6348defcb2db2d338c.jpg

Figure 2: Overview of PALU. (left) Token-level Unlearning. Comparison of how different methods (classic, current token-aware, and ours) distinguish between token roles to identify specific optimization targets. (right) Vocabulary-level Unlearning. Visualization of theoretical probability distributions induced by different objectives (PDU, -CE, and our local entropy maximization) relative to the original LLM prediction.

图 2：PALU 概览。(左) 词元级去学习。比较不同方法（经典方法、当前的词元感知方法和我们的方法）如何区分词元角色以识别特定的优化目标。(右) 词汇级去学习。可视化由不同目标（PDU、-CE 和我们的局部熵最大化）相对于原始 LLM 预测所诱导的理论概率分布。

therefore overestimates the scope of intervention and can unnecessarily degrade general language ability. To address this, we adopt a semantic-aware filtering strategy. Following TPO (Zhou et al., 2026), we employ language models (e.g., DistilBERT (Sanh et al., 2019) or GPT-4) to identify the spans containing sensitive information. Formally, for an output sequence y of length T, we define a binary sensitivity mask $m_t \in \{0, 1\}$, where $m_t = 1$ denotes that token y_t belongs to a sensitive span, and $m_t = 0$ indicates a common token.

因此，这会高估干预的范围，并可能不必要的降低一般语言能力。为了解决这个问题，我们采用了一种语义感知的过滤策略。遵循 TPO（Zhou 等，2026），我们使用语言模型（例如 DistilBERT（Sanh 等，2019）或 GPT-4）来识别包含敏感信息的片段。正式地，对于长度为 T 的输出序列 y，我们定义一个二进制敏感度掩码 $m_t \in \{0, 1\}$，其中 $m_t = 1$ 表示标记 y_t 属于一个敏感片段，而 $m_t = 0$ 表示一个普通标记。

We further refine this paradigm by exploiting the temporal sparsity of generation. Even among sensitive tokens, typically only the first few are pivotal in triggering the specific semantics, while subsequent tokens merely elaborate on the determined path. Therefore, we propose to intervene solely on the initiating sensitive tokens. Let $I_{sens} = \{t \mid m_t = 1\}$ be the set of indices for sensitive tokens. We define the subset of optimization targets $I_{init} \subset I_{sens}$ by selecting only the first N indices from each sensitive span. Consequently, the output tokens are partitioned into three roles for optimization:

我们进一步通过利用生成的时序稀疏性来完善这一范式。即使在敏感标记中，通常也只有前几个标记对于触发特定语义至关重要，而后续的标记只是对已确定路径的进一步阐述。因此，我们提出仅对起始敏感标记进行干预。设 $I_{sens} = \{t \mid m_t = 1\}$ 为敏感标记的索引集合。我们通过从每个敏感区间中仅选择前 N 个索引来定义优化目标的子集 $I_{init} \subset I_{sens}$。因此，输出标记被划分为三个角色进行优化：

(1) Initiating Targets ($t \in I_{init}$): These pivotal tokens are subjected to our unlearning objective to disrupt the sensitive trajectory.
(2) Common Tokens ($t \notin \mathcal{I}_{\text{sens}}$): These context-agnostic tokens are constrained by a KL divergence

(1) 启动目标 ($t \in I_{init}$)：这些关键标记会受到我们的去学习目标的影响，以干扰敏感轨迹。
(2) 常规标记 ($t \notin \mathcal{I}_{\text{sens}}$)：这些上下文无关的标记受到 KL 散度的约束。

loss to preserve general utility and fluency.

损失以保持一般用途和流畅性。

(3) Redundant Sensitive Tokens ($t \in \mathcal{I}_{\text{sens}} \quad \mathcal{I}_{\text{init}}$): The remaining sensitive tokens are excluded from the computation graph, adhering to our principle of minimal intervention.

(3) 冗余敏感令牌 ($t \in \mathcal{I}_{\text{sens}} \quad \mathcal{I}_{\text{init}}$)：剩余的敏感令牌被排除在计算图之外，遵循我们最小干预的原则。

4.3 Local Entropy Maximization

4.3 局部熵最大化

From an entropy perspective, the standard negated CE objective exhibits a fundamental limitation for unlearning. While it suppresses the probability of the target token, it fails to guarantee an increase in the entropy of the predictive distribution. This is because the suppressed probability mass may simply shift to another specific token (e.g., a highly correlated synonym), causing the distribution to retain a peaked (low-entropy) shape. Such uncontrolled probability redistribution prevents the model from achieving a state of true ignorance and often leads to unstable behavior (Jia et al., 2024).

从熵的角度来看，标准的否定 CE 目标函数在遗忘任务中存在根本性的限制。虽然它抑制了目标标记的概率，但无法保证预测分布的熵增加。这是因为被抑制的概率质量可能会简单地转移到另一个特定的标记（例如，高度相关的同义词），导致分布保持尖峰（低熵）的形状。这种不受控制的概率重新分布阻止了模型达到真正的无知状态，并常常导致不稳定的行为（Jia 等，2024）。

In contrast, prior work such as PDU (Entesari et al., 2025) offers a theoretically superior interpretation: effective unlearning should push the predictive distribution toward a maximum-entropy state. Formally, this involves maximizing $H(\mathbf{y}_t) = -\sum_{i=1}^{|\mathcal{V}|} p_i \log p_i$, where $|\mathcal{V}|$ is the vocabulary size, and p_i denotes the probability of token i . Maximum entropy is achieved when the distribution becomes uniform (i.e., $p_i = 1/|\mathcal{V}|$), implying total uncer-

相比之下，先前的工作（如 PDU，Entesari 等，2025）提供了一个理论上更优的解释：有效的遗忘应该将预测分布推向最大熵状态。形式上，这涉及最大化 $H(\mathbf{y}_t) = -\sum_{i=1}^{|\mathcal{V}|} p_i \log p_i$ ，其中 $|\mathcal{V}|$ 是词汇表的大小，而 p_i 表示标记 i 的概率。当分布变得均匀（即 $p_i = 1/|\mathcal{V}|$ ）时，达到最大熵，这意味着完全的不确定性。

tainty in the model's prediction and thus achieving thorough erasure. However, maximizing entropy over the entire vocabulary $\mathcal{V}$ is computationally prohibitive and practically unnecessary. We observe that autoregressive decoding in LLMs is dominated by a small set of high-probability candidates. Increasing the entropy of long-tail, low-probability tokens has a negligible impact on the outputs, yet consumes the majority of computational resources.

模型预测中的不确定性，从而实现彻底的擦除。然而，在整个词汇表上最大化熵 $\mathcal{V}$ 在计算上是不可行的，而且实际上也是不必要的。我们观察到，在大型语言模型中，自回归解码主要由一小部分高概率候选词主导。增加长尾、低概率标记的熵对输出几乎没有影响，却消耗了大部分计算资源。

To reconcile the structural robustness of maximum entropy with intervention efficiency, we propose local entropy maximization. We restrict the optimization scope to the decoding-critical subspace, aiming to maximize entropy within the top-K logits. This objective provides a stable local surrogate that encourages higher entropy among decoding-critical candidates, without optimizing the full-vocabulary entropy. For a target token $t \in I_{init}$, let $z_t \in R^{|\mathcal{V}|}$ be the logit vector. We define the set of indices for the top-K values as V_{top} . Crucially, to ensure optimization stability, V_{top} is identified using the frozen reference model $P_{\theta_{ref}}$ and remains fixed throughout the unlearning process. The localized objective is defined as:

为了在最大熵的结构鲁棒性与干预效率之间取得平衡，我们提出局部熵最大化方法。我们将优化范围限制在解码关键的子空间内，旨在最大化前 K 个 logit 的熵。该目标提供了一个稳定的局部替代目标，鼓励解码关键候选词具有更高的熵，而无需优化整个词汇表的熵。对于目标 token $t \in I_{init}$，设 $z_t \in R^{|\mathcal{V}|}$ 为 logit 向量。我们定义前 K 个值的索引集合为 V_{top}。关键的是，为确保优化的稳定性，V_{top} 使用冻结的参考模型 $P_{\theta_{ref}}$ 进行识别，并在整个去学习过程中保持固定。局部化的目标定义如下：

$$\mathcal{L}_{\text{local}}(z_t) = \frac{1}{K} \sum_{i \in \mathcal{V}_{\text{top}}} \left(z_{t,i} - c\right)^2 . \tag{3}$$

This objective minimizes the variance among the dominant logits by encouraging them to converge toward a target value c. As a result, it serves a dual role. First, by reducing pairwise discrepancies among the top-K logits, it flattens the decoding-critical subspace, which in turn promotes a higher-entropy, locally uniform distribution after softmax normalization. Second, the target value c acts as an anchoring mechanism: choosing a sufficiently small c further suppresses the aggregate probability mass of the top-K candidates relative to the rest of the vocabulary. The effect of different choices of c is examined empirically in Section 5.

该目标通过鼓励这些主导的 logits 收敛到一个目标值 c，从而最小化它们之间的方差。因此，它起到了双重作用。首先，通过减少前 K 个 logits 之间的成对差异，它使解码关键的子空间趋于平坦，从而在 softmax 归一化后促进一个更高熵值、局部均匀分布。其次，目标值 c 起到一个锚定机制的作用：选择一个足够小的 c 值可以进一步抑制前 K 个候选项相对于整个词汇表其余部分的累积概率质量。不同 c 值选择的效果在第 5 节中进行了实证分析。

Finally, combining the semantic-aware token selection from Section 4.2 with this localized objective, our total unlearning loss is formulated as:

最后，结合第 4.2 节中基于语义的 token 选择与该局部目标，我们的总遗忘损失公式为：

$$
\begin{aligned}
\mathcal{L}_f = & \mathbb{E}_{t\in\mathcal{J}_{\text{init}}}[\mathcal{L}_{\text{local}}(z_t)] \\
& +\lambda\mathbb{E}_{t\notin\mathcal{J}_{\text{sens}}}[\text{KL}(P_{\theta_{\text{ref}}}(\cdot|x,y_{<t})\|P_\theta(\cdot|x,y_{<t}))].
\end{aligned}
\tag{4}
$$

This formulation strictly aligns with our efficiency principle: gradients are computed only for the sparse set of initiating tokens (I_{init}) and common tokens (via KL), while the redundant sensitive tokens ($t \in \mathcal{J}_{\text{sens}} \quad \mathcal{J}_{\text{init}}$) naturally result in zero gradients at these specific positions.

这种公式严格遵循我们的效率原则：梯度仅计算用于初始的稀疏标记（I_{init} ）和常见标记（通过 KL 散度），而冗余的敏感标记（$t \in \mathcal{J}_{\text{sens}} \quad \mathcal{J}_{\text{init}}$ ）在这些特定位置自然产生零梯度。

4.4 Complexity Analyses

4.4 复杂度分析

Regarding the forget objective, negated CE based unlearning incurs a dense backward cost of $\mathcal{O}(T|\mathcal{V}|)$. Our method in Equation 4 induces sparse gradients only on N initiating important tokens and top-K vocabulary dimensions. Although initiating tokens may appear in multiple disjoint spans, the total number of such tokens is always bounded by the output length T. Consequently, the complexity of the unlearning operation is bounded by $\mathcal{O}(TK)$, which is strictly lower than $\mathcal{O}(T|\mathcal{V}|)$ for $K \ll |V|$.

关于遗忘目标，基于否定 CE 的去学习会带来一个密集的反向成本 $\mathcal{O}(T|\mathcal{V}|)$ 。我们的方法在公式 4 中仅在 N 个初始重要 token 和前 K 个词汇维度上诱导稀疏梯度。尽管初始 token 可能出现在多个不相交的片段中，但此类 token 的总数始终被输出长度 T 所限制。因此，去学习操作的复杂度被限制在 $\mathcal{O}(TK)$ ，这比 $\mathcal{O}(T|\mathcal{V}|)$ 在 $K \ll |V|$ 时严格更低。

5 Experiments

5 实验

5.1 Experiment Setup

5.1 实验设置

Benchmarks. We evaluate PALU on two complementary unlearning benchmarks: TOFU (Maini et al., 2024) and MUSE (Shi et al., 2025). TOFU (Maini et al., 2024) presents a synthetic dataset of 200 fictitious authors across three forgetting granularities (1%, 5%, and 10%). MUSE (Shi et al., 2025) evaluates verbatim memorization and privacy risks in real-world News and Books.

基准测试。我们在两个互补的遗忘基准测试上评估 PALU：TOFU（Maini 等人，2024）和 MUSE（Shi 等人，2025）。TOFU（Maini 等人，2024）提供了一个合成数据集，包含 200 位虚构作者，覆盖三种遗忘粒度（1%、5% 和 10%）。MUSE（Shi 等人，2025）则在现实世界中的新闻和书籍中评估逐字记忆和隐私风险。

Evaluation Metrics. To evaluate unlearning performance, we employ several representative metrics. For the TOFU dataset, we utilize Model Utility (MU), Forget Quality (FQ) (Maini et al., 2024), Fluency (Mekala et al., 2025), Exact Memorization (EM) (Tirumala et al., 2022) and Truth Ratio on D_f (F-TR), D_r (R-TR), real authors (Ra-TR), and real world knowledge (Rw-TR). Conversely, for the MUSE dataset, we adopt three metrics: Verbatim Memorization (VerbMem), Knowledge Memorization (KnowMem), and Privacy Leakage (PrivLeak) (Shi et al., 2025).

评估指标。为了评估去学习的性能，我们采用了一些具有代表性的指标。对于 TOFU 数据集，我们使用模型效用（MU）、遗忘质量（FQ）（Maini 等，2024）、流畅性（Mekala 等，2025）、精确记忆（EM）（Tirumala 等，2022）以及在 D_f 上的真实比率（F-TR）、D_r（R-TR）、真实作者（Ra-TR）和真实世界知识（Rw-TR）。相反，对于 MUSE 数据集，我们采用三个指标：逐字记忆（VerbMem）、知识记忆（KnowMem）和隐私泄露（PrivLeak）（Shi 等，2025）。

Baseline Methods. We evaluate the performance of PALU against a comprehensive set of baselines. First, we define two reference models: Original denotes the model trained on $D_f \cup D_r$, while Retain refers to the model trained exclusively on D_r , which serves as the ideal state for unlearning. Furthermore, we benchmark our method against various state-of-the-art (SOTA) approaches, including GA (Yao et al., 2024), GD (Yao et al., 2024), DPO (Rafailov et al., 2023), NPO (Zhang et al., 2024), SimNPO (Fan et al., 2025), PDU (Entesari et al., 2025) and TPO (Zhou et al., 2026).

基准方法。我们评估了 PALU 相对于一系列全面的基准方法的性能。首先，我们定义了两个参考模型：Original 表示在 $D_f \cup D_r$ 上训练的模型，而 Retain 表示仅在 D_r 上训练的模型，这作为去学习的理想状态。此外，我们还将我们的方法与各种最先进的（SOTA）方法进行了基准测试，包括 GA（Yao 等，2024）、GD（Yao 等，2024）、DPO（Rafailov 等，2023）、NPO（Zhang 等，2024）、SimNPO（Fan 等，2025）、PDU（Entesari 等，2025）和 TPO（Zhou 等，2026）。

Models. We conduct experiments on TOFU using

模型。我们在 TOFU 上进行实验使用

Method	Year	FQ (□)	MU (□)	Fluency (□)	EM (□)	F-TR (□)	Ra-TR (□)	R-TR (□)	Rw-TR (□)
Llama-2-7B									
Original	-	5.87E-14	0.6276	0.8557	0.9988	0.5113	0.6120	0.4596	0.5521
Retain	-	1.0000	0.6266	0.8889	0.6670	0.6696	0.6052	0.4639	0.5624
GA	2024	5.95E-11	0.5580	0.7423	0.9215	0.5304	0.5919	0.4608	0.5426
GD	2024	0.0396	0.3577	0.2334	0.6429	0.5839	0.5651	0.4497	0.5958
DPO	2023	0.5453	0.5503	0.6984	0.6155	0.6822	0.5138	0.4416	0.5051
NPO	2024	0.6284	0.5920	0.8115	0.6574	0.6623	0.6155	0.4613	0.5663
SimNPO	2025	0.4663	0.5921	0.9093	0.7343	0.6707	0.6437	0.4138	0.5776
PDU	2025	0.0021	0.5111	0.4834	0.6498	0.7600	0.6217	0.3490	0.6348
TPO	2026	0.6284	0.5862	0.7929	0.6621	0.6618	0.5907	0.4515	0.5967
PALU	2026	0.7126	0.6238	0.8122	0.5935	0.7030	0.6701	0.4762	0.6069
Llama-3.1-8B									
Original	-	6.54E-13	0.6276	0.8522	0.9978	0.4788	0.4963	0.5298	0.6218
Retain	-	1.0000	0.6323	0.8857	0.6167	0.6216	0.5256	0.5279	0.6127
GA	2024	8.05E-07	0.5838	0.8182	0.8281	0.5532	0.5279	0.4766	0.6196
GD	2024	0.2705	0.5536	0.8012	0.7153	0.6245	0.5333	0.4601	0.6069
DPO	2023	0.4663	0.5531	0.8761	0.6374	0.6320	0.5203	0.4794	0.5076
NPO	2024	0.6284	0.6006	0.8527	0.6803	0.6424	0.6226	0.4608	0.5801
SimNPO	2025	0.6284	0.5767	0.8626	0.6459	0.6514	0.7007	0.4726	0.5886

PDU	2025	0.5226	0.5705	0.8114	0.6621	0.6535	0.6031	0.4631	0.5968
TPO	2026	0.7216	0.5921	0.8571	0.5973	0.6522	0.6154	0.4638	0.7286
PALU	2026	0.9238	0.6162	0.8656	0.6518	0.6617	0.6455	0.4882	0.6447

方法	年	FQ (□)	MU (□)	流利度（□）	EM (□)	F-TR (□)	Ra-TR (□)	R-TR (□)
Llama-2-7B								
请提供需要翻译的文本内容。	-	5.87E-14	0.6276	0.8557	0.9988	0.5113	0.6120	0.4596
保留	-	1.0000	0.6266	0.8889	0.6670	0.6696	0.6052	0.4639
GA	2024	5.95E-11	0.5580	0.7423	0.9215	0.5304	0.5919	0.4608
GD	2024	0.0396	0.3577	0.2334	0.6429	0.5839	0.5651	0.4497
DPO	2023	0.5453	0.5503	0.6984	0.6155	0.6822	0.5138	0.4416
NPO	2024	0.6284	0.5920	0.8115	0.6574	0.6623	0.6155	0.4613
SimNPO	2025	0.4663	0.5921	0.9093	0.7343	0.6707	0.6437	0.4138
PDU	2025	0.0021	0.5111	0.4834	0.6498	0.7600	0.6217	0.3490
TPO	2026	0.6284	0.5862	0.7929	0.6621	0.6618	0.5907	0.4515
巴鲁（PALU）	2026	0.7126	0.6238	0.8122	0.5935	0.7030	0.6701	0.4762
Llama-3.1-8B								
请提供需要翻译的文本内容。	-	6.54E-13	0.6276	0.8522	0.9978	0.4788	0.4963	0.5298
保留	-	1.0000	0.6323	0.8857	0.6167	0.6216	0.5256	0.5279
GA	2024	8.05E-07	0.5838	0.8182	0.8281	0.5532	0.5279	0.4766
GD	2024	0.2705	0.5536	0.8012	0.7153	0.6245	0.5333	0.4601
DPO	2023	0.4663	0.5531	0.8761	0.6374	0.6320	0.5203	0.4794
NPO	2024	0.6284	0.6006	0.8527	0.6803	0.6424	0.6226	0.4608
SimNPO	2025	0.6284	0.5767	0.8626	0.6459	0.6514	0.7007	0.4726
PDU	2025	0.5226	0.5705	0.8114	0.6621	0.6535	0.6031	0.4631
TPO	2026	0.7216	0.5921	0.8571	0.5973	0.6522	0.6154	0.4638
帕卢	2026	0.9238	0.6162	0.8656	0.6518	0.6617	0.6455	0.4882

Table 1: Overall performance from forget 5% split on TOFU benchmark with different unlearning methods and models. We bold the best and underline the second-best.

表 1: 在 TOFU 基准上使用不同去学习方法和模型的 forget 5% 分割的整体表现。我们用粗体标出最佳结果，并用下划线标出次佳结果。

Llama-2-7B and Llama-3.1-8B, and on MUSE using Llama-2-7B (Dubey et al., 2024; Touvron et al., 2023) as primary backbones.

Llama-2-7B 和 Llama-3.1-8B，在 MUSE 中使用 Llama-2-7B（Dubey 等人，2024；Touvron 等人，2023）作为主要主干网络。

Further details are provided in the Appendix.

附录中提供了更多细节。

5.2 Overall Performance

5.2 总体性能

Due to page limitations, we focus our main analysis on TOFU, which offers fine-grained and controllable evaluation of forgetting behaviors through explicit QA dependencies. We additionally evaluate our method on the MUSE benchmark, which targets real-world memorization and privacy risks, and report consistent conclusions in Appendix F.3.

由于页面限制，我们主要分析 TOFU，它通过显式的问答依赖关系，提供了对遗忘行为的细粒度和可控的评估。我们还将在 MUSE 基准上评估我们的方法，该基准针对现实世界中的记忆和隐私风险，并在附录 F.3 中报告一致的结论。

Main Results on TOFU. Table 1 presents the performance of PALU on the forget 5% setting of the TOFU benchmark. Benefiting from its dual-locality mechanism, which integrates prefix truncation in the temporal dimension with logit flattening in the vocabulary dimension, our method achieves the best overall trade-off among the compared methods. In terms of FQ, PALU significantly outperforms the strongest baseline TPO, achieving relative improvements of 13.4% on Llama-2-7B and 28.0% on Llama-3.1. This widening gap in the more capable Llama-3.1 highlights the scalability of our approach in handling complex generation patterns. Crucially, regarding MU, PALU

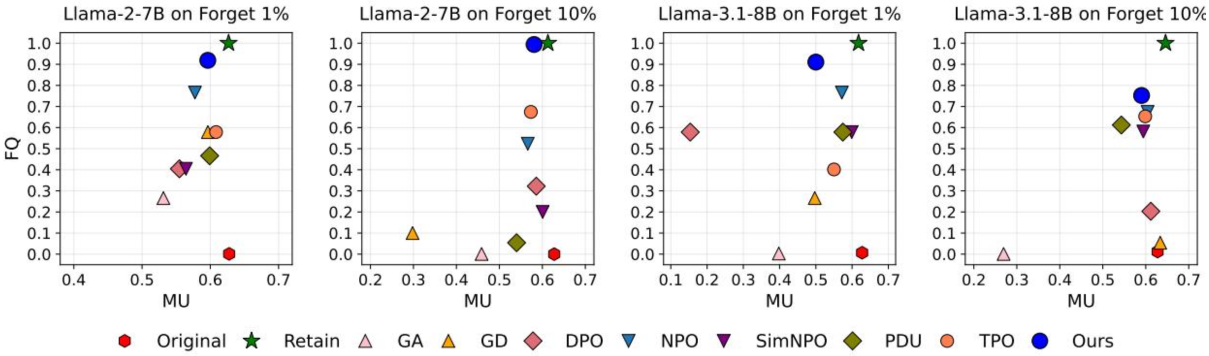
TOFU 主要结果。表 1 展示了 PALU 在 TOFU 基准的 forget 5% 设置中的表现。得益于其双局部性机制，该机制在时间维度上整合了前缀截断，在词汇维度上整合了 logit 扁平化，我们的方法在比较的方法中实现了最佳的整体权衡。在 FQ 方面，PALU 显著优于最强的基线 TPO，分别在 Llama-2-7B 和 Llama-3.1 上实现了 13.4% 和 28.0% 的相对提升。在更强大的 Llama-3.1 上，这一差距的扩大突显了我们的方法在处理复杂生成模式方面的可扩展性。关键的是，就 MU 而言，PALU

breaks the forgetting-utility trade-off commonly observed in prior works. GA and GD suffer from catastrophic collapse, while TPO and NPO sacrifice utility for forgetting. In contrast, PALU maintains an MU of 0.6238 on Llama-2-7B. This performance surpasses the score of TPO (0.5862) and virtually matches the Retain model reference score of 0.6266. This confirms that our localized intervention precisely targets sensitive knowledge without eroding the general capabilities of the model.

打破了以往研究中常见的遗忘与效用之间的权衡。GA 和 GD 会遭受灾难性崩溃，而 TPO 和 NPO 为了遗忘而牺牲了效用。相比之下，PALU 在 Llama-2-7B 上保持了 0.6238 的 MU。这一表现超过了 TPO（0.5862）的得分，并几乎与 Retain 模型的参考得分 0.6266 相匹配。这证实了我们的局部干预精确地针对了敏感知识，而不会削弱模型的一般能力。

Performance across Different Forget Ratios. Figure 3 illustrates the trade-off between MU and FQ on the forget 1% and 10% settings. Across both Llama-2-7B and Llama-3.1-8B, PALU demonstrates superior robustness, achieving a performance profile that aligns most closely with the Retain model compared to all baselines. In the demanding forget 10% setting, where competitive methods like NPO and DPO exhibit marked deterioration, PALU maintains stability and effectively replicates the ideal unlearning state. This confirms that PALU strikes the optimal balance between erasure and preservation, successfully circumventing the utility collapse seen in GA and GD.

不同遗忘比例下的性能表现。图 3 展示了在遗忘 1% 和 10% 设置下，MU 与 FQ 之间的权衡。在 Llama-2-7B 和 Llama-3.1-8B 模型上，PALU 展现出更优越的鲁棒性，其性能表现最接近保留模型，优于所有基线方法。在要求较高的遗忘 10% 设置下，像 NPO 和 DPO 这样的竞争方法表现出显著的性能下降，而 PALU 则保持稳定，并有效复制了理想的去学习状态。这证实了 PALU 在擦除与保留之间达到了最佳平衡，成功避免了 GA 和 GD 中出现的效用崩溃现象。



Original Retain GA GD DPO NPO SimNPO PDU TPO Ours

Original Retain GA GD DPO NPO SimNPO PDU TPO Ours

Figure 3: Performance on TOFU forget 1% and 10% split for different unlearning methods on different models.

图 3：不同模型上使用不同遗忘方法在 TOFU forget 1% 和 10% 分割下的性能表现。

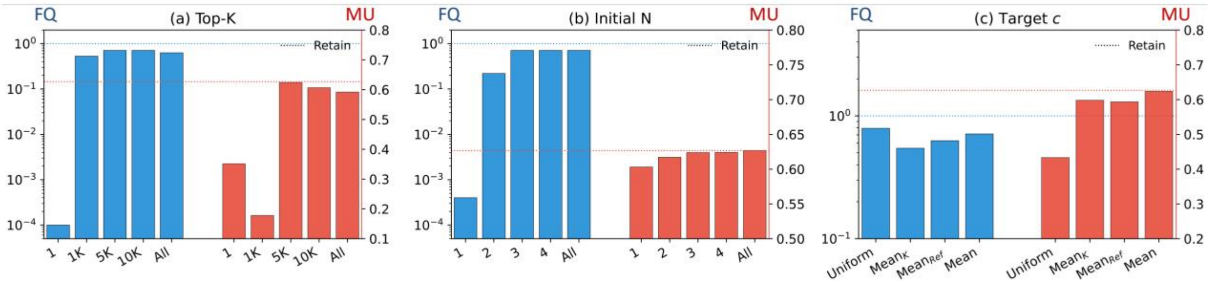


Figure 4: Analysis of PALU. We evaluate the impact of (left) the logit truncation size, (middle) the prefix length, and (right) the target threshold strategies, i.e., Uniform, Global Mean, and Local Mean. Blue bars represent Forget Quality (left y-axis), and red bars represent Model Utility (right y-axis).

图 4：PALU 分析。我们评估了以下因素的影响：（左）logit 截断大小，（中）前缀长度，以及（右）目标阈值策略，即均匀分布、全局均值和局部均值。蓝色条形图表示遗忘质量（左 y 轴），红色条形图表示模型效用（右 y 轴）。

5.3 Ablation Study: Validating Dual-Sparsity

5.3 消融实验：验证双稀疏性

To verify the effectiveness of our proposed dual-sparsity framework, we conduct ablation studies to isolate the contributions of Vocabulary Sparsity (via Top-K) and Temporal Sparsity (via Initial-N). When analyzing one component, we revert the other to its standard dense setting (i.e., full vocabulary or full sequence) to strictly evaluate the marginal gain of the specific module.

为了验证我们提出的双稀疏框架的有效性，我们进行了消融实验，以分离词汇稀疏性（通过 Top-K）和时间稀疏性（通过 Initial-N）的贡献。在分析其中一个组件时，我们将另一个组件恢复到其标准的密集设置（即完整词汇或完整序列），以便严格评估特定模块的边际增益。

Effect of Vocabulary Sparsity (Top-K). Figure 4(left) analyzes the impact of the logit truncation size K. We compare our sparse approach against the dense baseline (full vocabulary, equivalent to the scope of PDU). We observe a critical threshold effect: when K = 1, the MU suffers from catastrophic collapse, dropping to nearly 1×10^{-4} . This confirms that suppressing only the top-1 token is inherently unstable due to the semantic redundancy of LLMs, where probability mass easily shifts to synonyms. However, performance recovers rapidly as K increases, and notably, saturation occurs around K = 5,000. Comparing K = 5,000 with the “All” (Full Vocabulary) setting, the marginal gain in unlearning efficacy is negligible, yet the computational cost of the latter is significantly higher. This result validates our Vo-

词汇稀疏性的影响（Top-K)。图 4（左）分析了 logit 截断大小 K 的影响。我们将我们的稀疏方法与密集基线（完整词汇，相当于 PDU 的范围）进行比较。我们观察到一个关键的阈值效应：当 K=1 时，MU 遭受灾难性崩溃，下降到几乎 1×10^{-4} 。这证实了仅抑制 top-1 token 在语义冗余的 LLM 中本质上是不稳定的，因为概率质量很容易转移到同义词上。然而，随着 K 的增加，性能迅速恢复，并且值得注意的是，饱和发生在 K=5,000 附近。将 K=5,000 与 “所有”（完整词汇）设置进行比较，后者的去学习效果的边际增益可以忽略不计，而其计算成本却显著更高。这一结果验证了我们的 Vo-

cabulary Sparsity hypothesis: optimizing a critical subspace is sufficient to induce effective confusion, rendering the optimization of tail tokens redundant. Effect of Temporal Sparsity (Initial-N). Figure 4(middle) examines the necessity of unlearning the entire response versus truncating the gradient flow at the prefix. Optimizing a single token is too abrupt, thereby resulting in suboptimal utility. However, performance stabilizes at N = 3. Extending the optimization window beyond the first 3 tokens yields virtually zero additional gains in unlearning efficacy but linearly increases computational burden. This corroborates our temporal sparsity hypothesis: due to the autoregressive nature of LLMs, disrupting the entry point of a sensitive trajectory is sufficient to collapse the entire sequence. Thus, our prefix-based approach matches full-sequence efficacy while minimizing cost.

词汇稀疏性假设：优化一个关键子空间足以诱导有效的混淆，从而使尾部标记的优化变得冗余。时间稀疏性的影响（初始-N)。图 4（中间）考察了对整个响应进行去学习的必要性，与在前缀处截断梯度流动之间的差异。优化单个标记过于突然，从而导致效用次优。然而，性能在 N = 3 时趋于稳定。将优化窗口延伸到前 3 个标记之外，几乎不会带来额外的去学习效果，但会线性增加计算负担。这证实了我们的时序稀疏性假设：由于 LLM 具有自回归性质，破坏敏感轨迹的入口点就足以使整个序列崩溃。因此，我们的基于前缀的方法在保持完整序列效果的同时，最大限度地降低了成本。

5.4 Analysis of Optimization Target c

5.4 优化目标 c 的分析

Having established the optimal sparsity configurations ($K = 5,000$, $N = 3$), we further investigate the choice of the flattening target c in Equation 3. Figure 4(right) compares four strategies: uniform distribution (Uniform), the mean of Top- K logits (Mean $_K$), the mean of the reference model’s all

在确定了最优稀疏性配置（$K = 5,000$，$N = 3$）之后，我们进一步研究了方程 3 中展平目标 c 的选择。图 4（右）比较了四种策略：均匀分布（Uniform)、Top- K 逻辑值的平均值（Mean $_K$)、参考模型所有

Method	LOSS	ZLib	MinK	MinK++
Llama-2-7B				
Original	1.0000	1.0000	1.0000	1.0000
Retain	0.3568	0.3106	0.3513	0.4641
GA	0.9243	0.8202	0.9603	0.7103
GD	0.9243	0.8202	0.9603	0.7103
DPO	0.3614	0.2760	0.3658	0.8059
NPO	0.1734	0.1998	0.1740	0.1599
SimNPO	0.3176	0.2930	0.3065	0.3520
PDU	0.4729	0.5563	0.7090	0.8879

TPO	0.2459	0.1944	0.3857	0.7795
PALU	0.1840	0.1296	0.1643	0.1955
Llama-3.1-8B				
Original	1.0000	1.0000	1.0000	0.9999
Retain	0.3620	0.2958	0.3562	0.4880
GA	0.9422	0.9277	0.9434	0.8517
GD	0.6230	0.5591	0.6064	0.4316
DPO	0.2265	0.2262	0.1756	0.2603
NPO	0.5777	0.5110	0.5690	0.5162
SimNPO	0.2393	0.2020	0.2469	0.6362
PDU	0.4378	0.3889	0.4474	0.9103
TPO	0.3147	0.2796	0.2954	0.2269
PALU	0.1434	0.2379	0.1237	0.1400

方法	损失	ZLib	MinK	MinK++
Llama-2-7B				
原文	1.0000	1.0000	1.0000	1.0000
保留	0.3568	0.3106	0.3513	0.4641
GA	0.9243	0.8202	0.9603	0.7103
GD	0.9243	0.8202	0.9603	0.7103
DPO	0.3614	0.2760	0.3658	0.8059
NPO	0.1734	0.1998	0.1740	0.1599
SimNPO	0.3176	0.2930	0.3065	0.3520
PDU	0.4729	0.5563	0.7090	0.8879
TPO	0.2459	0.1944	0.3857	0.7795
巴鲁（PALU）	0.1840	0.1296	0.1643	0.1955
Llama-3.1-8B				
原文	1.0000	1.0000	1.0000	0.9999
保留	0.3620	0.2958	0.3562	0.4880
GA	0.9422	0.9277	0.9434	0.8517
GD	0.6230	0.5591	0.6064	0.4316
DPO	0.2265	0.2262	0.1756	0.2603
NPO	0.5777	0.5110	0.5690	0.5162
SimNPO	0.2393	0.2020	0.2469	0.6362
PDU	0.4378	0.3889	0.4474	0.9103
TPO	0.3147	0.2796	0.2954	0.2269
帕卢	0.1434	0.2379	0.1237	0.1400

Table 2: Extended metrics evaluated on different methods on the TOFU benchmark with forget ratio = 5%.

表 2: 在 TOFU 基准上，忘记比例为 5% 时不同方法评估的扩展指标。

logits (Mean $_{ref}$), and the global mean of the current model's all logits (Mean). The results indicate that the choice of c governs the trade-off between erasure depth and manifold preservation. The uniform target imposes a maximum entropy constraint that proves too strict, disrupting the logits' natural distribution. In contrast, the global mean strategy yields the most favorable balance. By pulling the sharp peaks of sensitive tokens down to the stable global average level, we effectively bury the sensitive signal into the background noise of the model. This approach removes the distinctiveness of the target without distorting the overall manifold structure. Consequently, we adopt the global mean as the standard objective to ensure stability.

logits（均值 $_{ref}$ ），以及当前模型所有 logits 的全局均值（均值）。结果表明，c 的选择决定了擦除深度与流形保持之间的权衡。统一的目标施加了一个最大熵约束，该约束过于严格，破坏了 logits 的自然分布。相比之下，全局均值策略实现了最有利的平衡。通过将敏感标记的尖峰拉低到稳定的全局平均水平，我们有效地将敏感信号埋入模型的背景噪声中。这种方法去除了目标的独特性，而不会扭曲整体流形结构。因此，我们采用全局均值作为标准目标以确保稳定性。

5.5 Training Efficiency and Convergence

5.5 训练效率和收敛性

While our method theoretically reduces complexity to $\mathcal{O}(TK)$, as analyzed in Section 4.4, Figure 5 confirms its practical efficiency. Compared to NPO, PALU exhibits aggressive convergence, saturating FQ by Epoch 5, whereas NPO requires nearly double the iterations due to a significant warm-up lag. Simultaneously, PALU alleviates deep utility degradation, recovering MU within just 2 epochs. This

虽然我们的方法在理论上将复杂度降低到 $\mathcal{O}(TK)$ ，如第 4.4 节的分析所示，图 5 证实了其在实践中的高效性。与 NPO 相比，PALU 表现出强烈的收敛性，在第 5 个 Epoch 时就已经饱和 FQ，而 NPO 由于显著的预热延迟，所需的迭代次数几乎是 PALU 的两倍。同时，PALU 缓解了深度效用的退化，在仅仅 2 个 Epoch 内就恢复了 MU。这

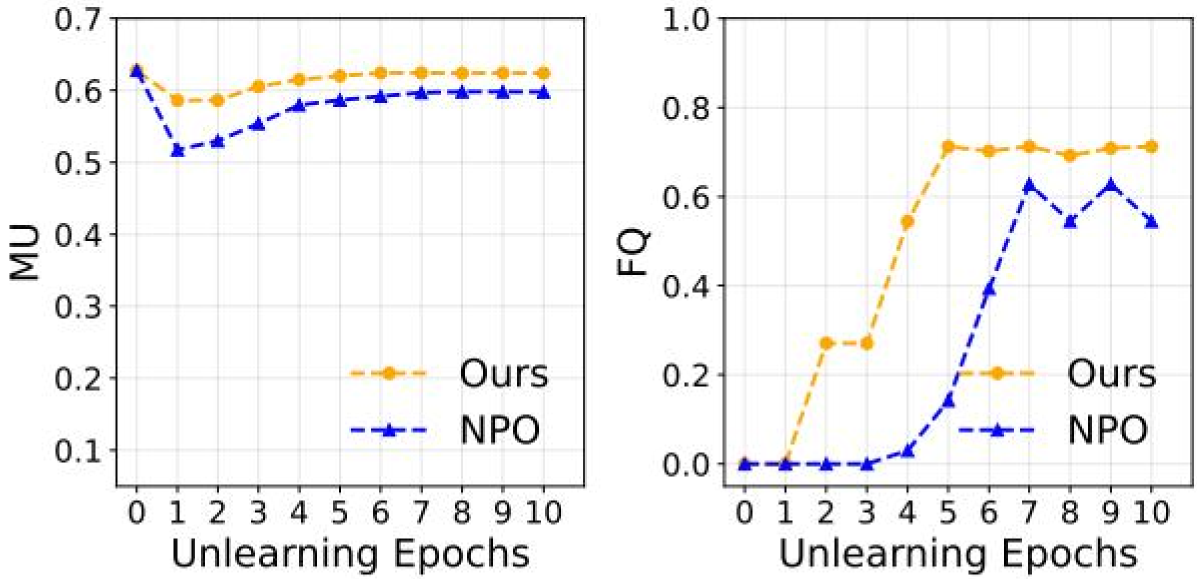


Figure 9: images/1b5a613b86465042a9b16f7fdec8e41c35aa76c43ae6ae7dcdffe850483c7f42.jpg

Figure 5: Convergence Analysis. Model Utility (MU) and Forget Quality (FQ) versus unlearning

epochs for PALU and NPO. The results are shown for the forget 5% split on the TOFU dataset over 10 epochs.

图 5：收敛性分析。模型效用（MU）和遗忘质量（FQ）与 PALU 和 NPO 的去学习轮次的关系。结果展示了在 TOFU 数据集上进行 10 轮次的 5% 遗忘分割后的结果。

Forget Ratio	FQ(□)	MU(□)
25%	0.1420	0.6291
50%	0.6872	0.5951
100%	0.6284	0.6086
Ours	0.7126	0.6238

忘记比率	FQ(□)	MU(□)
25%	0.1420	0.6291
50%	0.6872	0.5951
100%	0.6284	0.6086
我们的	0.7126	0.6238

Table 3: Performance comparing different prefix truncation ratios (25%, 50%, and 100%) of Llama-2-7B on the 5% forget split of the TOFU.

表 3: 在 TOFU 的 5% forget 分割上，比较 Llama-2-7B 不同前缀截断比例（25%、50% 和 100%）的性能。

rapid convergence effectively halves the required training duration, which, combined with our gradient sparsity, establishes PALU as a highly cost-effective solution for large-scale deployment.

快速收敛有效地将所需的训练时间减半，结合我们的梯度稀疏性，使 PALU 成为大规模部署的高性价比解决方案。

5.6 Performance Varying Splitting Ratios

5.6 性能随分割比例的变化

To evaluate the generalization of PALU across varying entity lengths, we conduct a sensitivity analysis shown in Table 3. The 50% ratio performs close to ours, suggesting that a moderate, length-aware budget can approximate the benefit of targeting the causally dominant tokens. Overall, dynamic ratios appear promising, but realizing consistent gains likely requires finer-grained hyperparameter search and better budget selection rules, which remain an important direction for future work.

为了评估 PALU 在不同实体长度下的泛化能力，我们进行了如表 3 所示的敏感性分析。50% 的比例表现接近我们的结果，这表明一个适中且具有长度感知的预算可以近似地获得针对因果主导标记的收益。总体而言，动态比例显示出良好的前景，但要实现一致的提升，可能需要更精细的超参数搜索和更好的预算选择规则，这仍然是未来工作的重要方向。

5.7 Performance on Other Metrics

5.7 其他指标的表现

Table 2 presents a rigorous evaluation using extended metrics: LOSS (Yeom et al., 2018), ZLib (Carlini et al., 2021), MinK (Shi et al., 2023), and MinK++ (Zhang et al., 2025b) on the TOFU benchmark. Unlike utility metrics, these likelihood-based indicators measure MIA separability between forget and holdout samples.

表 2 展示了使用扩展指标进行的严格评估：LOSS（Yeom 等人，2018）、ZLib（Carlini 等人，2021）、MinK（Shi 等人，2023）和 MinK++（Zhang 等人，2025b）在 TOFU 基准上的表现。与效用指标不同，这些基于似然的指标衡量了遗忘样本和保留样本之间的 MIA 可区分性。

PALU demonstrates strong target suppression, producing consistently low directional AUC values on Llama-2-7B and reducing the MinK++ directional AUC by 38.3% relative to TPO for Llama-

PALU 表现出强大的目标抑制能力，在 Llama-2-7B 上持续产生较低的方向性 AUC 值，并且相对于 TPO，将 Llama 的 MinK++ 方向性 AUC 减少了 38.3%

3.1-8B. Notably, PALU achieves a lower directional AUC than the Retain model, reducing the LOSS directional AUC to 0.1434 compared to 0.3620 on Llama-3.1-8B. This behavior is consistent with PALU prioritizing strong suppression of target knowledge over strictly matching the Retain model's distribution. Consequently, forget samples may receive lower target likelihoods than holdout samples representing ordinary unseen data.

3.1-8B. 显著地，PALU 在方向性 AUC 方面优于 Retain 模型，将 LOSS 方向性 AUC 降低至 0.1434，相比 Llama-3.1-8B 上的 0.3620。这种行为与 PALU 优先考虑对目标知识的强抑制，而非严格匹配 Retain 模型的分布是一致的。因此，遗忘样本可能比代表普通未见过数据的 holdout 样本获得更低的目标似然值。

6 Conclusion

6 结论

To overcome the limitations of indiscriminate token treatment in existing methods, we propose PALU, a framework driven by dual-sided localized entropy maximization. Our investigation reveals that effective unlearning does not require global suppression; instead, it can be achieved by precisely targeting the sensitive prefix in the temporal dimension and the top-K logits in the vocabulary dimension. This dual-locality mechanism allows PALU to sever sensitive generation paths with minimal computational overhead. Empirical results across diverse benchmarks demonstrate that PALU achieves the state-of-the-art overall trade-off, effectively erasing sensitive knowledge while robustly preserving the general utility of LLMs.

为了解决现有方法中对令牌无差别处理的局限性，我们提出了 PALU，一个由双侧局部熵最大化驱动的框架。我们的研究发现，有效的遗忘并不需要全局抑制；相反，可以通过在时间维度上精确针对敏感前缀，在词汇表维度上精确针对前 K 个 logits 来实现。这种双局部性机制使得 PALU 能够在计算开销最小的情况下切断敏感生成路径。在多种基准测试中实证结果表明，PALU 实现了目前最先进的整体权衡效果，有效擦除敏感知识，同时稳健地保留了大型语言模型的一般用途。

Acknowledgments

致谢

This work was supported by the advanced computing resources provided by the Supercomputing Center of USTC. We also acknowledge the GPU cluster built by the MCC Lab of the School of Information Science and Technology, USTC.

本研究得到了中国科学技术大学超级计算中心提供的先进计算资源的支持。我们还感谢中国科学技术大学信息科学与技术学院 MCC 实验室构建的 GPU 集群。

Limitations

限制

Currently, our framework is designed and validated exclusively on text-based LLMs. While PALU demonstrates superior efficacy in manipulating discrete textual token probabilities to unlearn sensitive concepts, it has not yet been extended to Multimodal Large Language Models (MLLMs) that integrate visual or audio modalities. Defining a sensitive prefix in a visual patch sequence or quantifying logit confusion in multi-modal vocabularies requires further investigation. We leave adapting dual-locality to multimodal privacy for future work.

目前，我们的框架仅针对基于文本的大型语言模型（LLMs）进行了设计和验证。虽然 PALU 在操纵离散文本标记概率以消除敏感概念方面表现出卓越的效果，但它尚未扩展到整合视觉或音频模态的多模态大型语言模型（MLLMs）。在视觉补丁序列中定义敏感前缀，或在多模态词汇表中量化 logit 混乱，仍需要进一步研究。我们将适应双局部性以实现多模态隐私作为未来的工作方向。

References

参考资料

Adam Berger, Stephen A Della Pietra, and Vincent J Della Pietra. 1996. A maximum entropy approach to

Adam Berger, Stephen A Della Pietra, 和 Vincent J Della Pietra. 1996. 一种最大熵方法用于

natural language processing. Computational linguistics, 22(1):39–71.

Nicholas Carlini, Florian Tramer, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlingsson, and 1 others. 2021. Extracting training data from large language models. In 30th USENIX security symposium (USENIX Security 21), pages 2633–2650.

Chen-Hao Chao, Chien Feng, Wei-Fang Sun, Cheng-Kuang Lee, Simon See, and Chun-Yi Lee. 2024. Maximum entropy reinforcement learning via energy-based normalizing flow. Advances in Neural Information Processing Systems, 37:56136–56165.

Jiaao Chen and Diyi Yang. 2023. Unlearn what you want to forget: Efficient unlearning for llms. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pages 12041–12052.

Daixuan Cheng, Shaohan Huang, Xuekai Zhu, Bo Dai, Xin Zhao, Zhenliang Zhang, and Furu Wei. 2026. Reasoning with exploration: An entropy perspective. In Proceedings of the AAAI Conference on Artificial Intelligence, volume 40, pages 30377–30385.

Vineeth Dorna, Anmol Mekala, Wenlong Zhao, Andrew McCallum, Zachary C Lipton, J Zico Kolter, and Pratyush Maini. 2025. OpenUnlearning: Accelerating LLM unlearning via unified benchmarking of methods and metrics. arXiv preprint arXiv:2506.12618.

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, and 1 others. 2024. The llama 3 herd of models. arXiv e-prints, pages arXiv–2407.

Taha Entesari, Arman Hatami, Rinat Khaziev, Anil Ramakrishna, and Mahyar Fazlyab. 2025. Constrained entropic unlearning: A primal-dual framework for large language models. In The Thirty-ninth Annual Conference on Neural Information Processing Systems.

Chongyu Fan, Jiancheng Liu, Licong Lin, Jinghan Jia, Ruiqi Zhang, Song Mei, and Sijia Liu. 2025. Simplicity prevails: Rethinking negative preference optimization for llm unlearning. In Neurips Safe Generative AI Workshop 2024.

Qingkai Fang, Yan Zhou, Shoutao Guo, Shaolei Zhang, and Yang Feng. 2025. Llama-omni2:

Llm-based real-time spoken chatbot with autoregressive streaming speech synthesis. arXiv preprint arXiv:2505.02625.
Xiaoliang Fu, Jiaye Lin, Yangyi Fang, Binbin Zheng, Chaowen Hu, Zekai Shao, Cong Qin, Lu Pan, Ke Zeng, and Xunliang Cai. 2026. Maspo: Unifying gradient utilization, probability mass, and signal reliability for robust and sample-efficient llm reasoning. arXiv preprint arXiv:2602.17550.
Jun Gao, Di He, Xu Tan, Tao Qin, Liwei Wang, and Tie-Yan Liu. 2019. Representation degeneration problem in training natural language generation models. arXiv preprint arXiv:1907.12009.
Anmol Goel, Alan Ritter, and Iryna Gurevych. 2026. Auditing language model unlearning via information decomposition. arXiv preprint arXiv:2601.15111.
Zhiyuan Han, Beier Zhu, Yanlong Xu, Peipei Song, and Xun Yang. 2025. Benchmarking and bridging emotion conflicts for multimodal emotion reasoning. In Proceedings of the 33rd ACM International Conference on Multimedia, pages 5528–5537.
Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2019. The curious case of neural text degeneration. arXiv preprint arXiv:1904.09751.
Jinpeng Hu, Tengteng Dong, Gang Luo, Hui Ma, Peng Zou, Xiao Sun, Dan Guo, Xun Yang, and Meng Wang. 2024. Psycollm: Enhancing llm for psychological understanding and evaluation. IEEE Transactions on Computational Social Systems, 12(2):539–551.
Edwin T Jaynes. 1957. Information theory and statistical mechanics. Physical review, 106(4):620.
Jiabao Ji, Yujian Liu, Yang Zhang, Gaowen Liu, Ramana R Kompella, Sijia Liu, and Shiyu Chang. 2024. Reversing the forget-retain objectives: An efficient llm unlearning framework from logit difference. Advances in Neural Information Processing Systems, 37:12581–12611.
Jinghan Jia, Yihua Zhang, Yimeng Zhang, Jiancheng Liu, Bharat Runwal, James Diffenderfer, Bhavya Kailkhura, and Sijia Liu. 2024. Soul: Unlocking the power of second-order optimization for llm unlearning. In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, pages 4276–4292.
Yuxian Jiang, Yafu Li, Guanxu Chen, Dongrui Liu, Yu Cheng, and Jing Shao. 2025. Rethinking entropy regularization in large reasoning models. arXiv preprint arXiv:2509.25133.
Antonia Karamolegkou, Jiaang Li, Li Zhou, and Anders Søgaard. 2023. Copyright violations and large language models. arXiv preprint arXiv:2310.13771.
Hang Li, Tianlong Xu, Kaiqi Yang, Yucheng Chu, Yanling Chen, Yichi Song, Qingsong Wen, and Hui Liu. 2025. Ask-before-detection: Identifying and mitigating conformity bias in llm-powered error detector for math word problem solutions. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 1597–1609.
Jilong Liu, Pengyang Shao, Wei Qin, Fei Liu, Yonghui Yang, and Richang Hong. 2026. Debate over mixed-knowledge: A robust multi-agent reasoning framework for incomplete knowledge graph question answering. In Proceedings of the AAAI Conference
on Artificial Intelligence, volume 40, pages 15333-15341.
Ruixuan Liu, Li Xiong, and 1 others. 2025a. Direct token optimization: A self-contained approach to large language model unlearning. arXiv preprint arXiv:2510.00125.
Sijia Liu, Yuanshun Yao, Jinghan Jia, Stephen Casper, Nathalie Baracaldo, Peter Hase, Yuguang Yao, Chris Yuhao Liu, Xiaojun Xu, Hang Li, and 1 others. 2025b. Rethinking machine unlearning for large language models. Nature Machine Intelligence, pages 1–14.
Ziyang Luo, Kaixin Li, Hongzhan Lin, Yuchen Tian, Mohan Kankanhalli, and Jing Ma. 2025. Tree-of-evolution: Tree-structured instruction evolution for code generation in large language models. In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 297–316.
Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary Chase Lipton, and J Zico Kolter. 2024. TOFU: A task of fictitious unlearning for LLMs. In First Conference on Language Modeling.
Anmol Reddy Mekala, Vineeth Dorna, Shreya Dubey, Abhishek Lalwani, David Koleczek, Mukund Rungta, Sadid A Hasan, and Elita AA Lobo. 2025. Alternate preference optimization for unlearning factual knowledge in large language models. In Proceedings of the 31st International Conference on Computational Linguistics, pages 3732–3752.
Eric Michaud, Ziming Liu, Uzay Girit, and Max Tegmark. 2023. The quantization model of neural scaling. Advances in Neural Information Processing Systems, 36:28699–28722.
Haowen Pan, Yixin Cao, Xiaozhi Wang, Xun Yang, and Meng Wang. 2024. Finding and editing multi-modal neurons in pre-trained transformers. In Findings of the Association for Computational Linguistics: ACL 2024, pages 1012–1037.
Haowen Pan, Xiaozhi Wang, Yixin Cao, Zenglin Shi, Xun Yang, Juanzi Li, and Meng Wang. 2025.

Precise localization of memories: A fine-grained neuron-level knowledge editing technique for LLMs. In The Thirteenth International Conference on Learning Representations.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. Advances in neural information processing systems, 36:53728–53741.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. arXiv preprint arXiv:1910.01108.

自然语言处理。计算语言学，22(1):39–71。

Nicholas Carlini, Florian Tramer, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlingsson, 和 1 位其他人。2021。从大型语言模型中提取训练数据。在第 30 届 USENIX 安全研讨会（USENIX Security 21）中，第 2633–2650 页。

Chen-Hao Chao, Chien Feng, Wei-Fang Sun, Cheng-Kuang Lee, Simon See, 和 Chun-Yi Lee。2024。通过能量归一化流实现最大熵强化学习。《神经信息处理系统进展》，37:56136–56165。

Jiaao Chen 和 Diyi Yang。2023。忘记你想忘记的：高效地从大型语言模型中删除知识。在第 2023 届经验方法自然语言处理会议论文集，第 12041–12052 页。

Daixuan Cheng, Shaohan Huang, Xuekai Zhu, Bo Dai, Xin Zhao, Zhenliang Zhang, 和 Furu Wei。2026。通过熵的观点进行探索推理。在第 40 届 AAAI 人工智能会议论文集，第 30377–30385 页。

Vineeth Dorna, Anmol Mekala, Wenlong Zhao, Andrew McCallum, Zachary C Lipton, J Zico Kolter, 和 Pratyush Maini。2025。OpenUnlearning：通过统一的方法和指标基准测试加速大型语言模型的遗忘。arXiv 预印本 arXiv:2506.12618。

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, 和 1 位其他人。2024。llama 3 模型群。arXiv e-prints，第 arXiv–2407 页。

Taha Entesari, Arman Hatami, Rinat Khaziev, Anil Ramakrishna, 和 Mahyar Fazlyab。2025。受限熵遗忘：大型语言模型的对偶框架。在第 39 届神经信息处理系统年会。

Chongyu Fan, Jiancheng Liu, Licong Lin, Jinghan Jia, Ruiqi Zhang, Song Mei, 和 Sijia Liu。2025。简单性优先：重新思考负偏好优化用于大型语言模型的遗忘。在 2024 年 Neurips 安全生成人工智能研讨会。

Qingkai Fang, Yan Zhou, Shoutao Guo, Shaolei Zhang, 和 Yang Feng。2025。Llama-omni2：基于大型语言模型的实时语音聊天机器人，具有自回归流式语音合成。arXiv 预印本 arXiv:2505.02625。

Xiaoliang Fu, Jiaye Lin, Yangyi Fang, Binbin Zheng, Chaowen Hu, Zekai Shao, Cong Qin, Lu Pan, Ke Zeng, 和 Xunliang Cai。2026。Maspo：统一梯度利用、概率质量和信号可靠性以实现鲁棒且样本高效的大型语言模型推理。arXiv 预印本 arXiv:2602.17550。

Jun Gao, Di He, Xu Tan, Tao Qin, Liwei Wang, 和 Tie-Yan Liu。2019。自然语言生成模型训练中的表示退化问题。arXiv 预印本 arXiv:1907.12009。

Anmol Goel, Alan Ritter, 和 Iryna Gurevych。2026。通过信息分解审计语言模型的遗忘。arXiv 预印本 arXiv:2601.15111。

Zhiyuan Han, Beier Zhu, Yanlong Xu, Peipei Song, 和 Xun Yang。2025。多模态情感推理中的基准测试与情感冲突的桥梁。在第 33 届 ACM 国际多媒体会议论文集，第 5528–5537 页。

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, 和 Yejin Choi。2019。神经文本退化的奇妙案例。arXiv 预印本 arXiv:1904.09751。

Jinpeng Hu, Tengteng Dong, Gang Luo, Hui Ma, Peng Zou, Xiao Sun, Dan Guo, Xun Yang, 和 Meng Wang。2024。Psycollm：增强大型语言模型的心理理解和评估。《计算社会系统 IEEE 交易》，12(2):539–551。

Edwin T Jaynes。1957。信息理论与统计力学。Physical review，106(4):620。

Jiabao Ji, Yujian Liu, Yang Zhang, Gaowen Liu, Ramana R Kompella, Sijia Liu, 和 Shiyu Chang。2024。反转遗忘-保留目标：从 logit 差异出发的高效大型语言模型遗忘框架。《神经信息处理系统进展》，37:12581–12611。

Jinghan Jia, Yihua Zhang, Yimeng Zhang, Jiancheng Liu, Bharat Runwal, James Diffenderfer, Bhavya Kailkhura, 和 Sijia Liu。2024。Soul：解锁二阶优化在大型语言模型遗忘中的力量。在第 2024 届经验方法自然语言处理会议论文集，第 4276–4292 页。

Yuxian Jiang, Yafu Li, Guanxu Chen, Dongrui Liu, Yu Cheng, 和 Jing Shao。2025。重新思考大型推理模型中的熵正则化。arXiv 预印本 arXiv:2509.25133。

Antonia Karamolegkou, Jiaang Li, Li Zhou, 和 Anders Søgaard。2023。版权侵犯与大型语言模型。arXiv 预印本 arXiv:2310.13771。

Hang Li, Tianlong Xu, Kaiqi Yang, Yucheng Chu, Yanling Chen, Yichi Song, Qingsong Wen, 和 Hui Liu。2025。询问后再检测：识别和缓解大型语言模型驱动的错误检测器在数学单词问题解决方案中的一

致性偏差。在第 63 届计算语言学协会年会（第 1 卷：长论文），第 1597–1609 页。
Jilong Liu, Pengyang Shao, Wei Qin, Fei Liu, Yonghui Yang, 和 Richang Hong。2026。混合知识的争论：用于不完整知识图问题回答的鲁棒多智能体推理框架。在第 40 届 AAAI 人工智能会议论文集，第 15333-15341 页。
Ruixuan Liu, Li Xiong, 和 1 位其他人。2025a。直接令牌优化：大型语言模型遗忘的自包含方法。arXiv 预印本 arXiv:2510.00125。
Sijia Liu, Yuanshun Yao, Jinghan Jia, Stephen Casper, Nathalie Baracaldo, Peter Hase, Yuguang Yao, Chris Yuhao Liu, Xiaojun Xu, Hang Li, 和 1 位其他人。2025b。重新思考大型语言模型的机器遗忘。《自然机器智能》，第 1–14 页。
Ziyang Luo, Kaixin Li, Hongzhan Lin, Yuchen Tian, Mohan Kankanhalli, 和 Jing Ma。2025。进化树：用于大型语言模型中代码生成的树结构指令进化。在第 63 届计算语言学协会年会（第 1 卷：长论文），第 297–316 页。
Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary Chase Lipton, 和 J Zico Kolter。2024。TOFU：用于大型语言模型的虚构遗忘任务。在第一届语言建模会议。
Anmol Reddy Mekala, Vineeth Dorna, Shreya Dubey, Abhishek Lalwani, David Koleczek, Mukund Rungta, Sadid A Hasan, 和 Elita AA Lobo。2025。用于大型语言模型中事实知识遗忘的替代偏好优化。在第 31 届计算语言学国际会议论文集，第 3732–3752 页。
Eric Michaud, Ziming Liu, Uzay Girit, 和 Max Tegmark。2023。神经扩展的量化模型。《神经信息处理系统进展》，36:28699–28722。
Haowen Pan, Yixin Cao, Xiaozhi Wang, Xun Yang, 和 Meng Wang。2024。在预训练变压器中查找和编辑多模态神经元。在计算语言学协会年会：ACL 2024 的发现，第 1012–1037 页。
Haowen Pan, Xiaozhi Wang, Yixin Cao, Zenglin Shi, Xun Yang, Juanzi Li, 和 Meng Wang。2025。精确定位记忆：用于大型语言模型的细粒度神经元级知识编辑技术。在第十三届学习表示国际会议。
Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, 和 Chelsea Finn。2023。直接偏好优化：你的语言模型实际上是一个奖励模型。《神经信息处理系统进展》，36:53728–53741。
Victor Sanh, Lysandre Debut, Julien Chaumond, 和 Thomas Wolf。2019。DistilBERT：BERT 的压缩版本：更小、更快、更便宜和更轻。arXiv 预印本 arXiv:1910.01108。